

Directed Graph Feedback Loop Stability & Memory Buffers

Real-Time DSP Safety Guarantees in Dynamic Modular Graphs

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Abstract

This document formulates the cycle detection algorithms and 1-block delay buffer history allocation mechanism ensuring 100% real-time audio thread stability in **Time Dilation DAW**.

1 Modular Audio Directed Graph Representation

The modular audio processing network is modeled as a Directed Graph $G = (V, E)$, where V represents processing nodes and $E \subseteq V \times V$ represents audio or control patch connections.

2 Feedback Loop Detection Algorithm

A feedback loop occurs when there exists a directed path $v_1 \rightarrow v_2 \rightarrow \dots \rightarrow v_k \rightarrow v_1$. Depth-First Search (DFS) with node visitation states $\text{State} \in \{\text{UNVISITED}, \text{VISITING}, \text{VISITED}\}$ identifies cyclic dependencies in $O(|V| + |E|)$ time complexity.

3 1-Block History Delay Buffer Solution

For any connection $e = (u, v) \in E$ forming a feedback loop cycle:

$$x_{\text{dest}}[n] = y_{\text{src, prevBlock}}[n], \quad \forall n \in [0, K_{\text{block}} - 1] \quad (1)$$

where $y_{\text{src, prevBlock}}$ is the audio buffer produced during block $N - 1$. This guarantees zero buffer starvation, zero thread deadlocks, and absolute DSP stability without requiring zero-delay matrix solvers.